

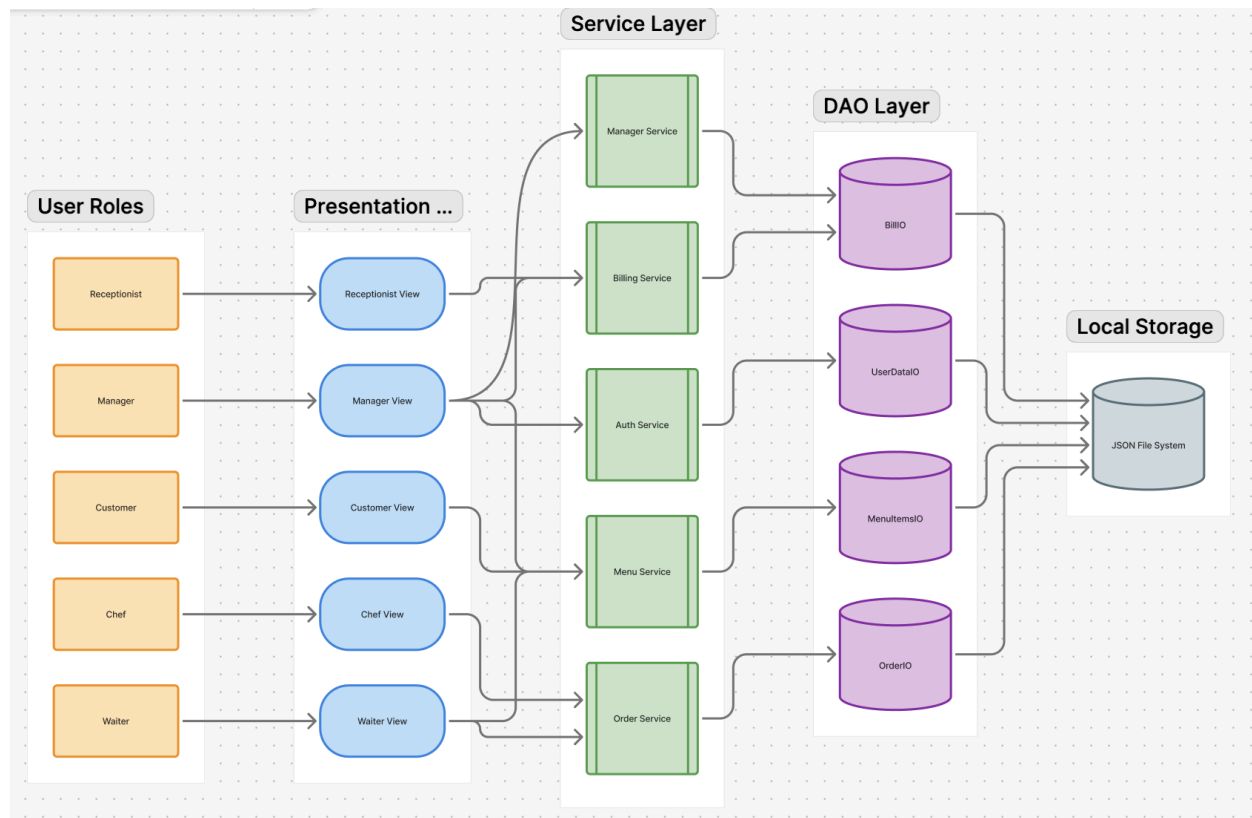
Restaurant Management System - Team B

PS:

A mid-sized restaurant faces operational challenges due to manual order processing, table booking, and billing procedures. These manual processes lead to delays, errors, and poor customer experience. To modernize operations, you are tasked with designing and developing a Java console-based Restaurant Management System that supports real-time updates and multiple concurrent users.

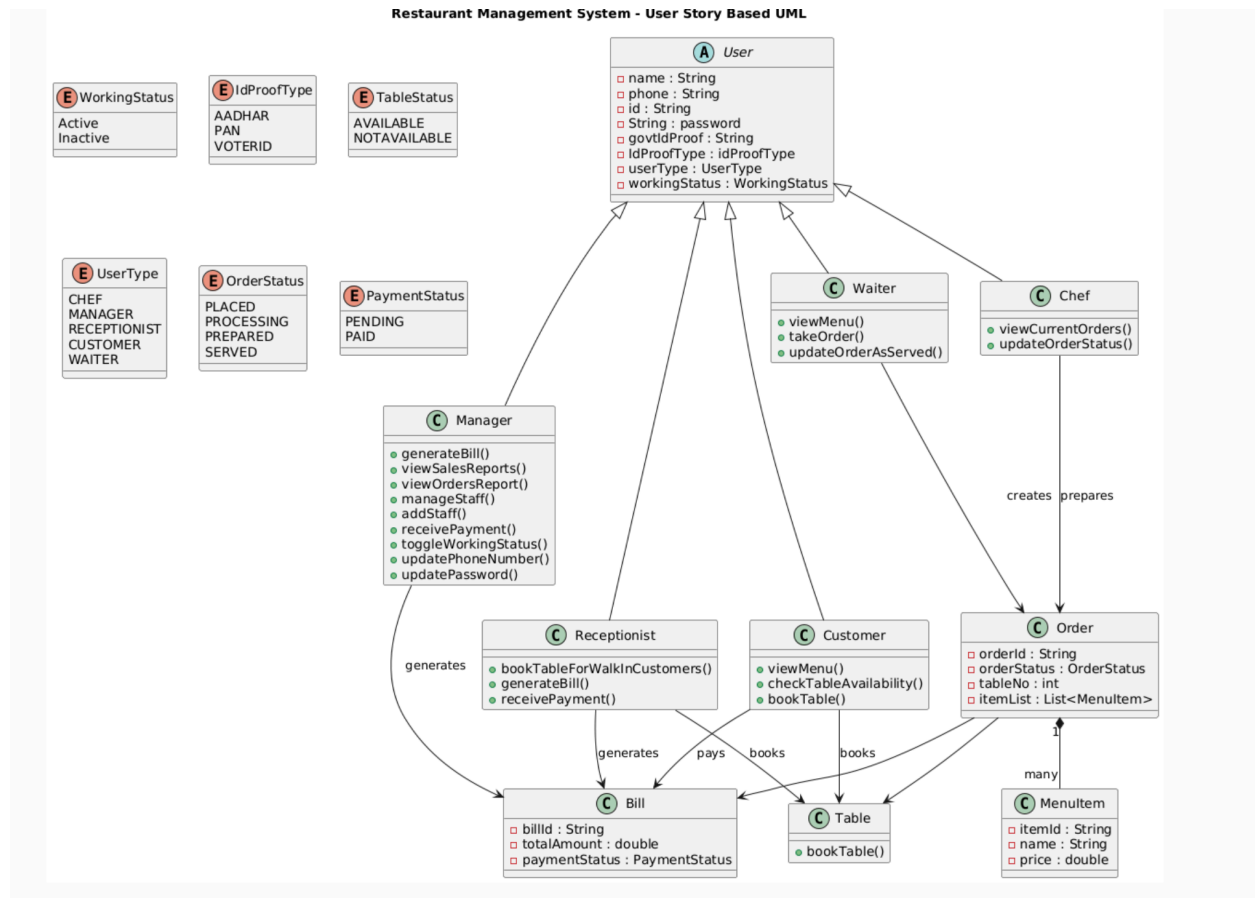
HLD:

High-Level Design: 3-tier architecture where the UI interacts with role-based services, which handle business logic and persist data through IO classes and JSON storage.



LLD:

Low-Level Design: Defines classes, models, methods, and interactions such as services coordinating with IO utilities and domain objects to execute booking, ordering, billing, and user operations.



Code Coverage Photo

Coverage AllTestsSuite x				
Element				
Class...	Method, %	Line, %	Branch, %	
services	100% (9/9)	100% (27/...)	85% (156/...	66% (60/...
MenuService	100% (1/1)	100% (1/1)	66% (4/6)	50% (2/4)
BookTableService	100% (1/1)	100% (1/1)	90% (10/11)	100% (4/4)
WaiterService	100% (1/1)	100% (3/3)	83% (15/18)	66% (4/6)
ManagerService	100% (1/1)	100% (9/9)	85% (63/74)	63% (23/...
ChefService	100% (1/1)	100% (4/4)	87% (14/16)	50% (3/6)
CustomerService	100% (1/1)	100% (3/3)	83% (15/18)	75% (6/8)
ReceptionistService	100% (1/1)	100% (3/3)	95% (19/20)	91% (11/12)
OrderService	100% (1/1)	100% (2/2)	80% (12/15)	50% (4/8)
AuthService	100% (1/1)	100% (1/1)	80% (4/5)	50% (3/6)

Welcome to Commit & Chutney

1. Login
2. Exit

CLI BASED
APPLICATION

LOGIN VIEW

Welcome to Commit & Chutney

1. Login

2. Exit

1

Enter User ID: 1

Enter Password: 123

Login Successful as MANAGER

Welcome Manager Sargam!

1. View Menu

2. Add Menu Item

3. Manage Staff

4. Add Staff

5. View Sales Reports

MANAGER DASHBOARD -

(Viewing the menu)

1. View Menu
2. Add Menu Item
3. Manage Staff
4. Add Staff
5. View Sales Reports
6. View Orders Reports
7. Generate Bill
8. Receive Payment
9. Logout

1

Item ID	Name	Price
1	Coffee	₹200.0
2	BisibeleBath	₹99.0
3	RagiMudde	₹39.0
4	Kichdi	₹49.0
5	ButterMilk	₹15.0

MANAGER - (ADDING MENU ITEM)

Welcome Manager Sargam!

1. View Menu
2. Add Menu Item
3. Manage Staff
4. Add Staff
5. View Sales Reports
6. View Orders Reports
7. Generate Bill
8. Receive Payment
9. Logout

2

Enter Item ID:

6

Enter Item Name:

Noodles

Enter Price:

80

Menu Item Added.

MANAGER - (Managing Staff data)-

Can change status and data

```
Manage Staff Data:
1. Change Working Status
2. Change staff data
2
Enter userId:
4
Change:
1. Phone number
2. Password
2
Enter Password:
123
Password updated.
```

MANAGER - (Generating Bill)-

```
Welcome Manager Sargam!

1. View Menu
2. Add Menu Item
3. Manage Staff
4. Add Staff
5. View Sales Reports
6. View Orders Reports
7. Generate Bill
8. Receive Payment
9. Logout
7
Enter Table Number (Order ID) for Bill:
1
Bill Generated: 299.00
```


MANAGER - can view order reports, receive payments, and then log out

9. Logout

8

Enter Table Number (Bill ID) for Payment:

1

Payment Received for billId : 1, ₹299.00

9. Logout

9

Logging out...

9. Logout

5

Sales Report:

Total Sales: 299.00

Amount Received : 0.00

Amount to be Received : 299.00

Waiter - login as waiter

Welcome to Commit & Chutney

1. Login

2. Exit

1

Enter User ID: 2

Enter Password: 123

Login Successful as WAITER

Welcome, Nirmith M R to Commit & Chutney|

1. View Menu

2. Take Order

3. Update order as Served

4. Logout

Waiter - taking an order through ordeID and tablenumber

Welcome, Nirmith M R to Commit & Chutney|

1. View Menu
2. Take Order
3. Update order as Served
4. Logout

2

Taking Order:

Enter Order ID, table number

1 1

Enter item ID and quantity:

1 1

Enter item ID and quantity:

2 1

Enter item ID and quantity:

-1 -1

Order has been placed.

Waiter - order is updates once served

Welcome, Nirmith M R to Commit & Chutney

1. View Menu
2. Take Order
3. Update order as Served
4. Logout

3

Update Order as Served:

Enter order id

1

Order served for table.

Waiter - logout

Welcome, Nirmith M R to Commit & Chutney

1. View Menu
2. Take Order
3. Update order as Served
4. Logout

4

Logging out...

1. Login
2. Exit

Chef - login

1. Login

2. Exit

1

Enter User ID: 3

Enter Password: 123

Login Successful as CHEF

Welcome, Divya to Commit & Chutney

Enter your choice:

1. View Current orders

2. Update order status

3. Logout

Chef - can view order

```
Welcome, Divya to Commit & Chutney
```

```
Enter your choice:
```

1. View Current orders
2. Update order status
3. Logout

```
1
```

```
Order Id | Order Status | ItemName
```

```
1    PROCESSING    BisibeleBath
```

```
1    PROCESSING    RagiMudde
```

Chef - update order status once prepared

Welcome, Divya to Commit & Chutney

Enter your choice:

1. View Current orders
2. Update order status
3. Logout

2

Enter Order id:

1

Order has been updated.

Chef - logout

```
Welcome, Divya to Commit & Chutney
```

```
Enter your choice:
```

1. View Current orders
2. Update order status
3. Logout

```
3
```

```
Logging out...
```

RECEPTIONIST DASHBOARD

1. Login

2. Exit

1

Enter User ID: 4

Enter Password: 123

Login Successful as RECEPTIONIST

Welcome, Rahul to Commit & Chutney

1. Book Table

2. Generate Bill

3. Recieve Payment

4. Logout

RECEPTIONIST - books table if customer walks in

Welcome, Rahul to Commit & Chutney

1. Book Table
2. Generate Bill
3. Recieve Payment
4. Logout

1

Booking table for walk-in customer...

Your Table is booked

Your Table number is : 3

RECEPTIONIST- generates bill for them

```
Welcome, Rahul to Commit & Chutney
```

1. Book Table
2. Generate Bill
3. Recieve Payment
4. Logout

```
2
```

```
Enter Table Number for Bill:
```

```
1
```

```
Bill generated successfully for table 1
```

RECEPTIONIST - logout

Welcome, Rahul to Commit & Chutney

1. Book Table
2. Generate Bill
3. Recieve Payment
4. Logout

4

Logging out...

1. Login
2. Exit

Customer - can view the menu , check for table availability and book table

1. Login

2. Exit

1

Enter User ID: 5

Enter Password: 123

Login Successful as CUSTOMER

Welcome, Krishna to Commit & Chutney

1. View Menu

2. Check Availability of Tables

3. Book a Table

4. Logout

Welcome, Krishna to Commit & Chutney

1. View Menu
2. Check Availability of Tables
3. Book a Table
4. Logout

1

Item ID	Name	Price
---------	------	-------

1	Coffee	₹200.0
2	BisibeleBath	₹99.0
3	RagiMudde	₹39.0
4	Kichdi	₹49.0
5	ButterMilk	₹15.0
6	Noodles	₹80.0

Welcome, Krishna to Commit & Chutney

1. View Menu
2. Check Availability of Tables
3. Book a Table
4. Logout

3

Booking table.

Your Table is booked

Your Table number is : 4

Table availability as seen by customer:

Welcome, Krishna to Commit & Chutney

1. View Menu
2. Check Availability of Tables
3. Book a Table
4. Logout

2

Table Availability:

Table 1 : UNAVAILABLE
Table 2 : UNAVAILABLE
Table 3 : UNAVAILABLE
Table 4 : UNAVAILABLE
Table 5 : AVAILABLE
Table 6 : AVAILABLE
Table 7 : AVAILABLE
Table 8 : AVAILABLE
Table 9 : AVAILABLE
Table 10 : AVAILABLE

CUSTOMER - logout

Welcome, Krishna to Commit & Chutney

1. View Menu
2. Check Availability of Tables
3. Book a Table
4. Logout

4

Logging out...

1. Login
2. Exit